



Target Operation Guides

Air Source Outdoor Unit				
Component	Abbreviation		Cooling Mode	Comments
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature
		Range	14~140Hz	
Outdoor Fan Motor	OFM	Operation	Target high pressure control (default: 355.6psig)	Fans speed control is used to maintain head pressure
		Range	0-39 step (0-1300rpm)	Typical Head Pressure range 355 psig - 469 psig
Main EEV	E_M	Operation	Full Open	
		Range	2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	SC = High Side Saturation Temp. - Liquid Line Temp.
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 100.4°F or all compressors in unit are at 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_a < 97°F and all compressors in unit are at 70Hz↑	Discharge Temperature Protection, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	On (Open)	Energized in cooling only
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 5-36 drgrees above outside air temperature	Used to calculate condenser sub-cooling
Mode Control Unit (MCU)				
Component	Abbreviation		Cooling Mode	Comments
Port Heating Solenoid	Heat	Operation	Off (Closed)	
Port Cooling Solenoid	Cool	Operation	On (Open)	On with a call for cooling
Port EEV	EEV	Operation	Open 480 Steps	Open when a zone calls for cooling
		Range	0 - 480 Steps	
Sub-Cooling EEV	SC_EEV	Operation	Closed 0 Steps	Active only in Heating Main and Cooling Main
		Range	480 Steps	
Sub-Cooler In Temperature		Operation		
Sub-Cooler Out Temperature		Operation	Close to Suction Header Temperature	
Indoor Unit IDU				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	

Air Source Outdoor Unit				
Component	Abbreviation		Cooling Main Mode	Comments
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	Primary function, Maintain Average Evaporator Temperature
		Range	14~140Hz	
Outdoor Fan Motor	OFM	Operation	Target high pressure control (default: 398 psig)	Fans speed control is used to maintain head pressure
		Range	0-39 step (0-1300rpm)	Typical Head Pressure Range 355 psig – 469 psig
Main EEV	E_M	Operation	Cooling & Heating capacity balance control	
		Range	300 - 2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F) / Discharge Temp. Protection	
		Range	0-480 step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 100.4°F or all compressors in unit are at 50Hz↓	Activates Sub-Cooling Control
			Off (Close): T_a < 96.8°F and all compressors in unit are at 70Hz↑	Discharge Temperature Control, Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	Off (Closed)	Helps force hot gas to the MCU & IDU's
Main Cooling Solenoid	VMC	Operation	On (Open)	
Condenser Out Thermistor	T_CO	Value	Should be 5-36 drgrees above outside air temperature	Used to calculate condenser sub-cooling
Mode Control Unit (MCU)				
Component	Abbreviation		Cooling Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open) Call for Heating Off (Closed) Call for Cooling	
Port Cooling Solenoid	Cool	Operation	On (Open) Call for Cooling Off (Closed) Call for Heat	
Port EEV	EEV	Operation	Open 480 Steps for Cooling and 0 Steps heating Operation	Open 480 step when a zone calls for cooling
		Range	0 - 480 Steps	Closed 0 Steps when a zone is in heating opens on mode change
Sub-Cooling EEV	SC_EEV	Operation	Active in Cooling Main	Provide subcooling for cooling zones
		Range	Modulating	Maintains 5.4 degrees of superheat between pipe in and out temps.
Sub-Cooler In Temperature		Operation	Should be colder then Pipe Out	
Sub-Cooler Out Temperature		Operation	Should be hotter then pipe in.	
Indoor Unit IDU Cooling Mode				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	65~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp (Typical range 0-7 degrees)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
		Operation	Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	
Indoor Unit IDU Heating Mode				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
		Operation	Medium speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	



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Air Source Outdoor Unit				
Component	Abbreviation		Heating Mode	Comments
Compressor	Comp	Operation	Target High pressure control (default: 427 psig)	Primary function is to maintain head pressure
		Range	20~160Hz	
Outdoor Fan Motor	OFM	Operation	Low Pressure Control	Low Side saturation temperature is "typically" 2-10 degrees lower than outside air temperature
		Range	0-39 step (0-1300rpm)	
Main EEV	E_M	Operation	*EVI Superheat control (default : SH 3.6°F -9°F)	Maintains evaorator superheat in the heating mode.
		Range	125 - 2000 Steps	
EVI EEV	E_EVI	Operation	On (Open)	
		Range	0-480step	
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 59°F or all compressors in unit are at 50Hz↓	Discharge temperature protection
			Off (Close): T_a < 50°F and all compresors in unit are at 70Hz↑	Low Ambient heating Operation and Flash/ Vapor Injection Circuit controlled by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	Off (Closed)	
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the outside air temperature	Typically it ranges between 2 to 10 degrees below outside air temperature
Mode Control Unit (MCU)				
Component	Abbreviation		Heating Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open)	On when in heating mode
Port Cooling Solenoid	Cool	Operation	Off (Closed)	
Port EEV	EEV	Operation	Closed 0 Steps	Open when a zone satisfies
		Range	0 - 480 Steps	
Sub-cooling EEV	SC_EEV	Operation	Closed 0 Steps	Active only in Heating Main and Cooling Main
		Range	480 Steps	
Sub-Cooler In Temperature		Operation		
Sub-Cooler Out Temperature		Operation	Close to Suction Header Temperature	
Indoor Unit IDU				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	



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		Range	20~160Hz	
Outdoor Fan Motor	OFM	Operation	Low pressure control	Low Side saturation temperature is "typically" 2-10 degrees lower than outside air temperature
		Range	0-39 step (0-1300rpm)	
Main EEV	E_M	Operation	*EVI Superheat control (default : SH 3.6°F or 9°F)	Modulates to maintain 3.6 or 9 degrees of evaporator superheat
		Range	125 - 2000 Steps	
EVI EEV	E_EVI	Operation	On (Open)	Discharge protecion or Flash-vapor injection
		Range	0-480step	
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 59°F or all compressor in unit are at 50Hz↓ Off (Close): T_a < 50°F and all compresors in unit are at 70Hz↑	Low Ambient heating Operation and discharge temperature protection. Control by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compresor is running)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Main EEV Solenoid Valve (HR Only)	V_ME	Operation	Off (Closed)	
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the outside air temperature	Typically it ranges between 2 to 10 degrees below outside air temperature
Mode Control Unit (MCU)				
Component	Abbreviation		Heating Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open) Call for Heating Off (Closed) Call for Cooling	
Port Cooling Solenoid	Cool	Operation	On (Open) Call for Cooling Off (Closed) Call for Heat	
Port EEV	EEV	Operation	Closed 0 Steps	Open when a zone satisfies or for mode change
		Range	0 - 480 Steps	
Sub-Cooling EEV	SC_EEV	Operation	Active in heating Main	Provide subcooling for cooling zones
		Range	Modulating	Maintains 5.4 degrees of superheat between pipe in and out temps.
Sub-Cooler In Temperature		Operation	Should be colder then Pipe Out	
Sub-Cooler Out Temperature		Operation	Should be hotter then pipe in.	
Indoor Unit IDU				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
		Operation	Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	
Indoor Unit IDU Cooling Mode				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = Pipe out Temp - Pipe in Temp (Typical range 0-7 degrees)
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1400 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
		Operation	Medium speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	